

A **constructor** in JavaScript is a special mechanism used to **create and initialize object instances**. JavaScript offers two main approaches to defining constructors: Modern ES6 Classes and Traditional Constructor Functions.

<https://www.geeksforgeeks.org/javascript/js-constructor-method/>

It sets up object properties and is typically invoked using the `new` keyword. Constructors allow for the creation of multiple instances with similar properties and methods.

```
var bmw1 = {  
  model: "X5",  
  color: "Black",  
  year: 2024  
};
```

```
var bmw2 = {  
  model: "M4",  
  color: "Blue",  
  year: 2023  
};
```

```
console.log(bmw1);  
console.log(bmw2);
```

Output

```
{ model: 'X5', color: 'Black', year: 2024 }  
{ model: 'M4', color: 'Blue', year: 2023 }
```

Problem: You have to repeat the same property names every time.

```
var BMW = function(model, color, year) {  
    this.model = model;  
    this.color = color;  
    this.year = year;  
};
```

```
var car1 = new BMW("X5", "Black", 2024);  
var car2 = new BMW("M4", "Blue", 2023);
```

```
console.log(car1);  
console.log(car2);
```

Output

```
BMW { model: 'X5', color: 'Black', year: 2024 }  
BMW { model: 'M4', color: 'Blue', year: 2023 }
```

constructor with method

```
function BMW(model, color, year) {  
    this.model = model;  
    this.color = color;  
    this.year = year;  
  
    this.display = function () {  
        console.log(this.model + " " + this.color + "  
" + this.year);  
    };  
}
```

```
var car1 = new BMW("X5", "Black", 2024);  
var car2 = new BMW("M4", "Blue", 2023);
```

```
car1.display();  
car2.display();
```

```

// 2 object literals
var winstonTeen = {
  "nickname": "Winsteen",
  "age": 15,
  "x": 20,
  "y": 50
};
var winstonAdult = {
  "nickname": "Mr. Winst-a-lot",
  "age": 30,
  "x": 229,
  "y": 50
};

var drawWinston = function(winston) {
  fill(255, 0, 0);
  var img = getImage("creatures/Winston");
  image(img, winston.x, winston.y);
  var txt = winston.nickname + ", " + winston.age;
  text(txt, winston.x+20, winston.y-10);
};

```

```

var Winston = function(nickname, age, x, y) {
  this.nickname = nickname;
  this.age = age + "yrs old";
  this.x = x;
  this.y = y;
};

var winstonTeen = new Winston("Winsteen", 15, 20, 50);
var winstonAdult = new Winston("Mr. Winst-a-lot", 30, 229, 50);

var drawWinston = function(winston) {
  fill(255, 0, 0);
  var img = getImage("creatures/Winston");
  image(img, winston.x, winston.y);
  var txt = winston.nickname + ", " + winston.age;
  text(txt, winston.x+20, winston.y-10);
};

```

Without Constructor

Create each object manually

Code repetition

Suitable for a few objects

No common blueprint

With Constructor

Create objects using **new**

Reusable code

Suitable for many objects

Common blueprint for all objects

Syntax

Without constructor:

 JavaScript



```
var obj = {  
  property1: value1,  
  property2: value2  
};
```

With constructor:

 JavaScript



```
function ConstructorName(property1, property2) {  
  this.property1 = property1;  
  this.property2 = property2;  
}  
  
var obj = new ConstructorName(value1, value2);
```

